

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology
University Examination May 2011
CE202 Data Communication and Networking

Date: 18.05.11, Wednesday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1

- (a) Explain the four fundamental characteristics of Data Communications Systems. 4
- (b) What are the advantages of a multipoint connection over a point-to-point connection? 2
- (c) List the responsibilities of Transport Layer. Briefly explain each. 5

Q.2

- (a) List three techniques of digital-to-digital conversion. Explain different Line Coding Schemes. 8
- (b) How can a composite signal be decomposed into its individual frequencies? What does the Nyquist theorem have to do with communications? 4

OR

Q.2

- (a) Explain the need of Modulation in the field of communication. Compare and contrast PCM and DM. 6
- (b) Describe The phases requires for actual communication in a circuit-switched network. Give the difference between circuit-switched network and packet switched network. 6

Q.3

- (a) Describe the goals of multiplexing. Distinguish between a link and a channel in multiplexing. Which of the three multiplexing techniques is (are) used to combine analog signals and digital signals? 6
- (b) What are the three major classes of guided media? How do guided media differ from unguided media? What is the significance of the twisting in twisted-pair cable? 6

OR

Q.3

- (a) Write a short note on Public Switched Telecom Network (PSTN). 6
- (b) How does a single-bit error differ from a burst error? Discuss the concept of redundancy in error detection and correction. Distinguish between forward errors corrections versus error correction by retransmission. 6

SECTION – II

Q.4

- (a) Write a short note on null modem. 6
- (b) Differentiate between error control and flow control. How they are useful in communication? 5

Q.5

- (a) Write a difference between following: 6
1. Serial Vs Parallel.
2. Asynchronous Transmission Vs Synchronous Transmission.
- (b) List the various address used at Data Link Layer, Internet Layer And Transport Layer in networking. 6

OR

Q.5

- (a) Explain frequency division multiplexing. 6
(b) What is redundancy check? Explain various methods which are using redundancy in Data Link Layer frame. 6

Q.6 Write a short note on following (Any TWO). 12

- (a) Go-Back-N Automatic Repeat Request.
(b) TCP/IP Protocol Suite.
(c) WAN.